

The empirical recurrence for OEIS A318094, recovered from its tail

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Abstract

OEIS A318094 records “Empirical recurrence of order 72” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 96$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{72} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-72 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A318094 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 1, 2, 3 or 5 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

2, 50, 220, 1563, 9193, 57744, 356786, 2215533, ...

The entry states:

Empirical recurrence of order 72 (see link above)

As of the “Last modified” line on the live entry (Aug 16 2018 10:28:59, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 96$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 96$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 72: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 72.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 192$ exact terms from $n = 2$ on, it returns order 72 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	1	c_{19}	1408611	c_{37}	−18518153	c_{55}	−321471
c_2	34	c_{20}	2228571	c_{38}	−14553091	c_{56}	−13897
c_3	50	c_{21}	−3659166	c_{39}	−31458558	c_{57}	16555
c_4	−267	c_{22}	−8550334	c_{40}	−334304	c_{58}	−58416
c_5	−778	c_{23}	7657271	c_{41}	29126769	c_{59}	108328
c_6	103	c_{24}	17145533	c_{42}	12185557	c_{60}	−17958
c_7	2884	c_{25}	−6978017	c_{43}	11870680	c_{61}	−33743
c_8	3917	c_{26}	−12159802	c_{44}	−30611346	c_{62}	−812
c_9	1159	c_{27}	−10337035	c_{45}	−9823094	c_{63}	15063
c_{10}	−2717	c_{28}	−13127996	c_{46}	24018684	c_{64}	−2666
c_{11}	−15389	c_{29}	44917487	c_{47}	2208693	c_{65}	−1361
c_{12}	−45286	c_{30}	42540041	c_{48}	−16131872	c_{66}	−162
c_{13}	−9312	c_{31}	−56167031	c_{49}	6383913	c_{67}	213
c_{14}	181844	c_{32}	−44662613	c_{50}	5863624	c_{68}	−8
c_{15}	187479	c_{33}	26709888	c_{51}	−5803459	c_{69}	60
c_{16}	−349353	c_{34}	7643118	c_{52}	−699444	c_{70}	−48
c_{17}	−666034	c_{35}	12297729	c_{53}	2002795	c_{71}	12
c_{18}	−103264	c_{36}	21652566	c_{54}	−51240	c_{72}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A318094 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{72} - c_1 x^{72-1} - \dots - c_{72}$. Its constant term is $-c_{72} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 72 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 72, so $\deg q = 72$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 72, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 72, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-72 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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